

Siun Kim

Post-doctoral Researcher
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EDUCATION

Mar. 2018 ~ Feb. 2023	Seoul National University Graduate School of Convergence Science and Technology Thesis: Clinical Information Extraction from Unstructured Free- Text for Pharmacovigilance <i>Advisor: Howard Lee</i> <i>Ph.D. in Medicine</i> GPA: 3.95 / 4.3	Seoul, Korea
Mar. 2013 ~ Feb. 2018	Seoul National University College of Liberal Studies Thesis: (Biology) The Consideration of the Relationship between Cellular Senescence and Organismal Aging / (Anthropology)Institutional Care Work and Daily Life of Elderly with Dementia: Focusing on Interaction with Caregivers in Adult Day Care Centers <i>Advisor: CHANHEE KANG</i> <i>B.S. in Liberal Studies: Anthropology and Biology (Summa Cum laude)</i> GPA: 3.98 / 4.3	Seoul, Korea

EXPERIENCE

- Post-doctoral Researcher at Seoul National University Hospital Biomedical Research Institute (March 2023 ~ July 2024, Adviser: Hyung-Jin Yoon)
- Research Assistant Professor at Seoul National University Hospital Biomedical Research Institute (August 2024 ~ Present)

RESEARCH INTERESTS

- Biomedical Natural Language Processing
- Machine Learning-based Drug Development
- Clinical Informatics
- Clinical Pharmacology & Regulatory Science

PUBLICATIONS (SCIE/ESCI)

1. Junmo Kim, Hyun-Lim Yang, Su Hwan Kim, Siun Kim, Jisoo Lee, Jiwon Ryu, Kwangsoo Kim, Zio Kim, Gun Ahn, Doyun Kwon, Hyung-Jin Yoon, "Deep learning-based long-term risk evaluation of incident type 2 diabetes using electrocardiogram in a non-diabetic population: a retrospective, multicentre study", *EClinicalMedicine*, (2024)
2. Siun Kim, Taegwan Kang, Tae Kyu Chung, Yoona Choi, YeSol Hong, Kyomin Jung, Howard Lee, "Extraction of Comprehensive Drug Safety Information from Adverse Drug Event Narratives in Spontaneous Reporting System", *Drug Safety*, (2023): 1-15.
3. Siun Kim, Yoona Choi, Jung-Hyun Won, Jung Mi Oh, Howard Lee, "An annotated corpus from biomedical articles to construct a drug-food interaction database", *JOURNAL OF BIOMEDICAL INFORMATICS*, (2022)
4. Siun Kim, Hyun A Lee, Seong Bok Jang, Howard Lee, "A population pharmacokinetic-pharmacodynamic model of YH12852, a highly selective 5-hydroxytryptamine 4 receptor agonist, in healthy subjects and patients with functional constipation", *CPT: PHARMACOMETRICS & SYSTEMS PHARMACOLOGY*, (2021)
5. Soohyun Kim, Siun Kim, Howard Lee, "A critical review of the United States regulatory pathways for determining the equivalence of efficacy between CT-P13 and original infliximab (Remicade®).", *DRUG DESIGN DEVELOPMENT AND THERAPY*, (2020)
6. Jung-Hyun Won, Yesol Hong, Siun Kim, Howard Lee, "One-year post-acute COVID-19 syndrome and mortality in South Korea: a nationwide matched cohort study using claims data", *FRONTIERS IN PUBLIC HEALTH*, (2024)
7. Siun Kim, David Seung U Lee, Yujin Kim, Hyung-Jin Yoon, Howard Lee, "Beyond Fine-Tuning: Leveraging Domain-Aware In-Context Learning with Large Language Models for Clinical Named Entity Recognition", (*UNDER REVIEW*), (2025)
8. Siun Kim, Jiwon Ryu, Sejoong Kim, Su Hwan Kim, Myeongju Kim, Hyung-Jin Yoon, "Improving Real-Time Prediction of Intradialytic Hypotension through Recurrence Pattern Integration and Evaluation Refinement", (*UNDER REVIEW*), (2025)
9. Chaiho Shin, Dareen Eom, Hahyun You, Kwangsoo Kim, Siun Kim*, Hyung-Jin Yoon*, "Contextual Span Detection with Open-Source Large Language Models Enhances Biomedical Entity Linking in Gastric Cancer Surgical Notes", (*UNDER REVIEW, Co-corresponding*), (2025)
10. Yujin Park, Siun Kim, Hyung-Jin Yoon, "Eating Alone as a Possible Mediator Between Living Alone and Depressive Symptoms: A Causal Mediation Analysis of KNHANES Data", (*UNDER REVIEW, Co-first*), (2025)

PUBLICATIONS (CONFERENCE)

1. Siun Kim, Hyung-Jin Yoon, "Questioning Our Questions: How Well Do Medical QA Benchmarks Evaluate Clinical Capabilities of Language Models?", BioNLP Workshop at ACL 2025, Vienna (July 2025) - Oral
2. Siun Kim, Jung-Hyun Won, David Lee, Renqian Luo, Lijun Wu, Tao Qin, Howard Lee, "CRese: Benchmark Data and Automatic Evaluation Framework for Recommending Eligibility Criteria from Clinical Trial Information", Findings of the Association for Computational Linguistics: EACL 2024 (Findings of EACL), March 2024, Malta
3. Siun Kim, Jung-Hyun Won, David Seung U Lee, Renqian Luo, Lijun Wu, Yingce Xia, Tao Qin, Howard Lee, "Revisiting Machine-Learning based Drug Repurposing: Drug Indications Are Not a Right Prediction Target", In Conference on Health, Inference, and Learning. PMLR, 2023.
4. Siun Kim, Hyung-Jin Yoon, "DiZiNER: Disagreement-guided Instruction Refinement via Simulating Pilot Annotation for Zero-shot Named Entity Recognition" (Under review)

PUBLICATIONS (BOOKS)

- Siun Kim, Howard Lee (2019). *The Era of Biologics is Coming*. Young Doc Press.

PROJECTS

- Development of an Intra-Dialysis Hypotension (IDH) Prediction Model using Time-Series Vital Sign and Dialyzer Data from Therapy Data Management System (TDMS), Seoul National University Hospital, Korea / [Leading Researcher] Overseeing and Performing All Data Preprocessing, Model Development, and Pap, Seoul National University Hospital, Korea / [Leading researcher] Oversee and perform all data preprocessing, model development, and paper writin (Mar. 2023 ~ Present)
- Collaboration research with Microsoft Research Asia: Revisiting and reformulating machine learning-based drug repositioning (DR) using clinical trials database and eligibility criteria information, Microsoft Research Asia, Korea / [Project leading] Reformulating ML-based DR study and engineering text prompts for DR (May. 2022 ~ Jun. 2023)
- Development of a prospective real-world registry for pediatric acute lymphoblastic leukemia (ALL) using multi-institutional electronic medical records (EHRs), Korea Health Industry Development Institute, Korea / Developing an extraction pipeline that captures key clinical information from CDW EHR data (Mar. 2022 ~ Feb. 2023)
- Development of natural language processing algorithms for extracting key clinical information from text data of drug adverse events reporting, Ministry of Food and Drug Safety., Korea / [Project leading] Performing task formulation and developing annotation guideline & annotated corpus (Feb. 2021 ~ Feb. 2022)
- Research on drug interaction evaluation using big data and machine learning, Ministry of Food and Drug Safety, Korea / [Subpart leading] Developing NLP models extracting drug-food interaction from biomedical texts (Feb. 2018 ~ Feb. 2022)
- Bacterial Exosome-based Anticancer Therapy: Immunooncology Trigger, Rosetta Exosome, Korea / [Leading Researcher] Developing TPP (Target Product Profile) of REX101 (Mar. 2021 ~ Dec. 2021)
- IDX-1197 development strategy consultation, Idience Co., Ltd, Korea / [Leading researcher] Constructing pre-clinical and clinical databases for approved PARP inhibitors (Oct. 2020 ~ Feb. 2021)

CONFERENCES

1. Siun Kim, "How to address data scarcity and bias for AI-powered drug discovery", Korean Society of Medical Informatics Conference, Seoul, Korea (December 2023) - Oral
2. Siun Kim, Taegwan Kang, TaeKyu Jung, Yoona Choi, Yesol Hong, Kyomin Jung, Howard Lee, "An Annotated Corpus and Natural Language Processing Model for Drug Safety Information Extraction from Free-texts Reporting Adverse Events", Korean Society of Medical Informatics Conference, Seoul, Korea (Nov. 2021) - Oral
3. Siun Kim, Howard Lee, Building Natural Language Training Data for Clinical Information Extraction [Tutorial Presentation], 2021 The Korean Society of Medical Informatics, Yongin, Korea (Nov. 2021)
4. Siun Kim, Hyun A Lee, Seong Bok Jang, Howard Lee, "A Population PK-PD Model of YH12852, a Highly Selective 5-HT₄ Receptor Agonist, in Healthy Subjects and Patients with Functional Constipation", Population Approach Group in Korea, Iksan, Korea (Nov. 2020) - Oral
5. Siun Kim, Hyun A Lee, Seol Ju Moon, Hyounggyoon Yoo, Seong Bok Jang, Howard Lee, "A Population Pharmacokinetic-Pharmacodynamic Model to Characterize the Increase in Gastrointestinal Motility by YH12852, a Highly Selective 5-HT₄ Receptor Agonist, in Healthy Subjects and Patients with Functional Constipation", 2019 AAPS PharmSci 360, San Antonio,

United States (Nov. 2019) - Poster

6. Siun Kim, Tae Kyu Jung, BongKi Moon, Jung Mi Oh, Howard Lee, "Development of an Annotation Guideline to Extract Drug-Food Interaction Information for Machine Learning", Korean Society for Clinical Pharmacology and Therapeutics, Yeosu, Korea (Nov. 2019) - Poster

SKILLS AND TECHNIQUES

- Programming
 - Python: PyTorch, NumPy, Pandas, SciKit-Learn
 - R
 - SQL
 - NONMEM (PK-PD simulation tool)
- Language
 - Korean (native)
 - English (business fluent)

AWARDS AND HONORS

- The Best Original Research Article, 1st, The Korean Society of Medical Informatics, Korea (Oct. 2021)
- The Best Article Presentation, 1st, Korean Society for Clinical Pharmacology and Therapeutics, Korea (Nov. 2021)
- Presidential Science Scholarship, Korea Student Aid Foundation, Korea (Mar. 2013 ~ Feb. 2017)

REVIEW EXPERIENCE

- ML4H 2024, ML4H 2025, BioNLP WS 2025

PATENTS

1. Howard Lee, Siun Kim, "Apparatus and method for extracting drug-food interaction information", KR-Application No. 10-2021-0172058